

# PocketSmart – Main Functionality Code Routing (Core Functionalities Development)

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## 1. Topic

Writing the main functionality code and routing for PocketSmart

Epic: Core Functionalities Development

## 2. Project Overview

PocketSmart is an AI-powered app that helps users track spending, set budgets, and get personalized money-saving suggestions. The main functions and routes turn user requests into actions such as saving an expense, checking a budget, or generating AI insights.

## 3. Objective

To write the main functionality code and connect it to the FastAPI routes, so each request from the user reaches the right function and returns a clear, correct response.

## 4. Work Steps

- 1 List the core features needed, such as expense tracking, budget setting, spending analysis, and AI insights.
- 2 Write a separate function for each feature, keeping the logic clean and reusable.
- 3 Create the routes using APIRouter and group them by feature.
- 4 Connect each route to its matching function and to the database session.
- 5 Use Pydantic models to validate the data coming into each route.
- 6 Connect the AI insight function to the Gemini model and return its suggestions through a route.
- 7 Set the correct HTTP methods, response models, and status codes for every route.
- 8 Add error handling so each route returns a clear message when something goes wrong.
- 9 Register all routers in the main application file.
- 10 Test each route from the FastAPI docs page and confirm the full flow works.

## 5. Main Parts of the Code Routing

- Core feature functions
- Router creation and grouping
- Database connection in routes
- Data validation with Pydantic

- AI insight route
- Response handling and error handling
- Router registration and testing